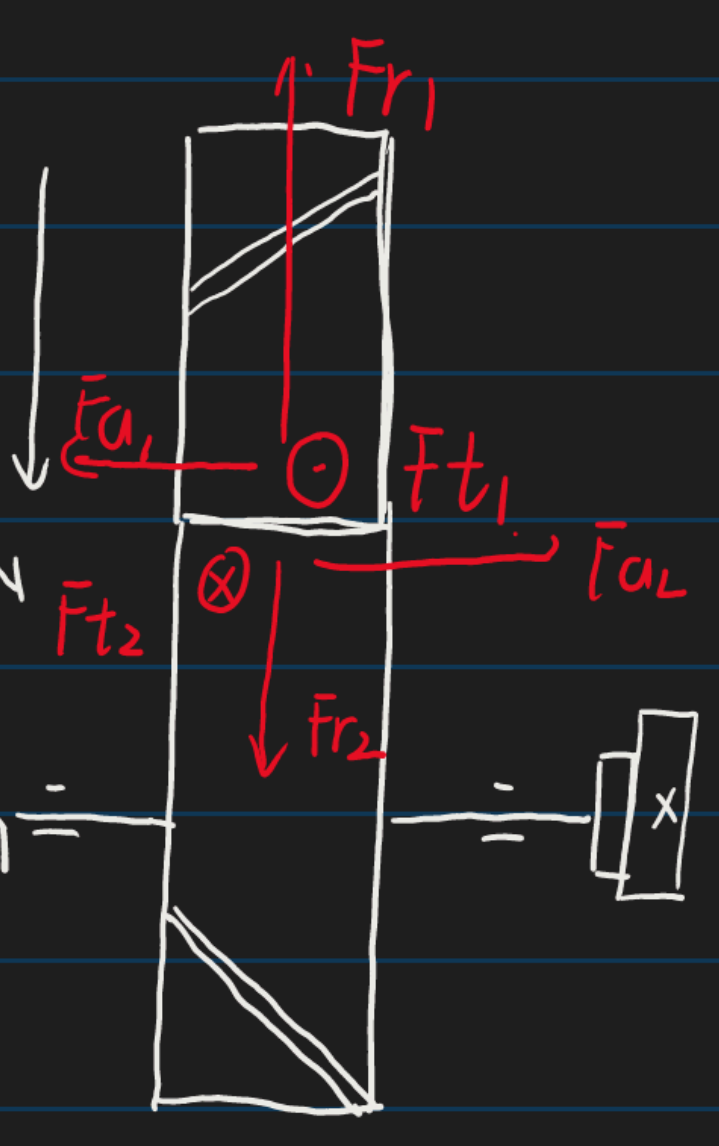
ZY-1

2025年12月3日 20:41

①使用45钢 调质处理



Fr1
n1
Fa1
Ft1
Fa2
Ft2
Fr2

P=4000W. n=750r/min

$P_{out}=T\cdot n/2\pi$ $T=\frac{P}{n}=9550\times\frac{4}{130}=294m\cdot N$

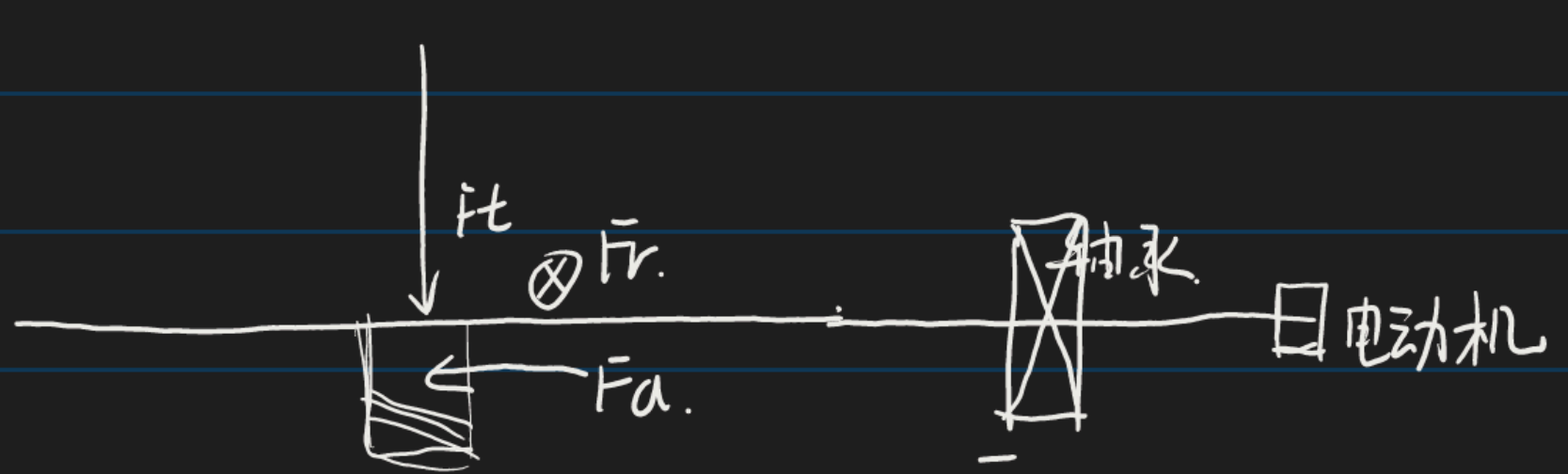
α=20° β=12° $d_1=0.3m$ B=90

$F_{t1}=\frac{2T}{d_1}=\frac{2\times294}{0.3}=F_{t2}=1960N\cdot m$

$F_{r1}=F_{t1}\cdot\tan\alpha_n/\cos\beta=\frac{294\cdot\tan20°}{\cos12°}=722N$

$F_{a1}=F_{t1}\tan\beta=1960\cdot\tan12°=417N$

$F_{n1}=F_{t1}/\cos\alpha_t\cdot\cos\beta_b$ X



Ft
Fr.
Fa.
轴承
电动机

②最小直径: 式15-2. $d\geq A_0\sqrt[3]{\frac{P}{n}}$

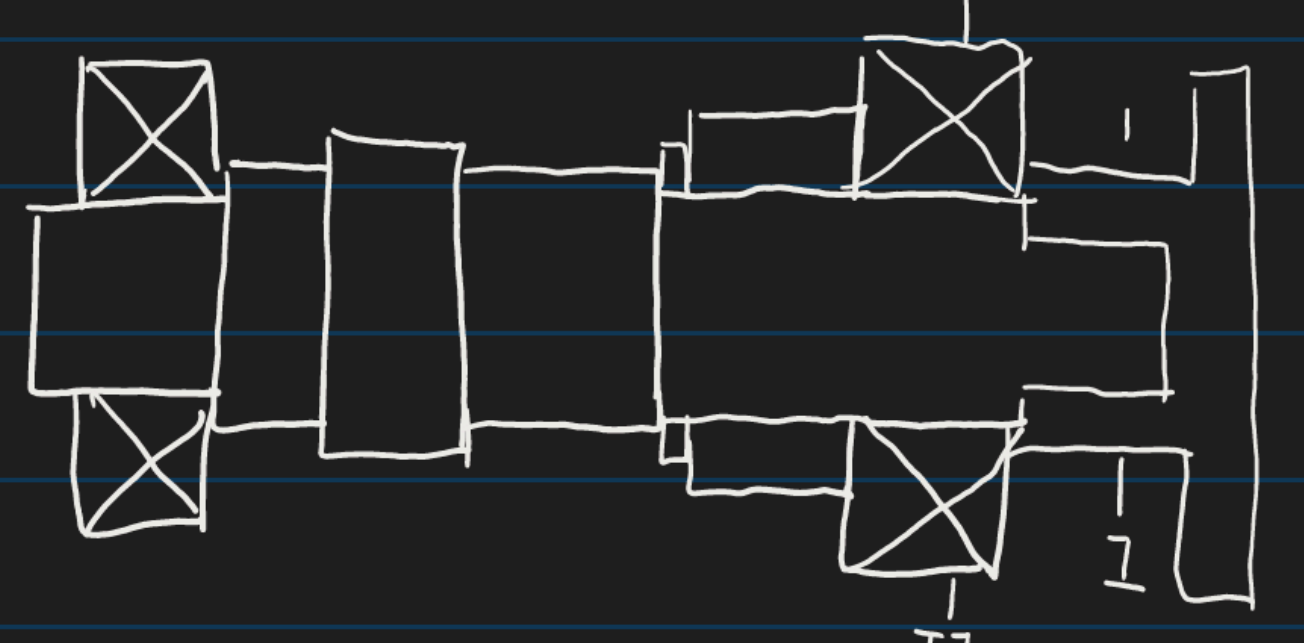
$A_0\rightarrow$ 材料 = 112

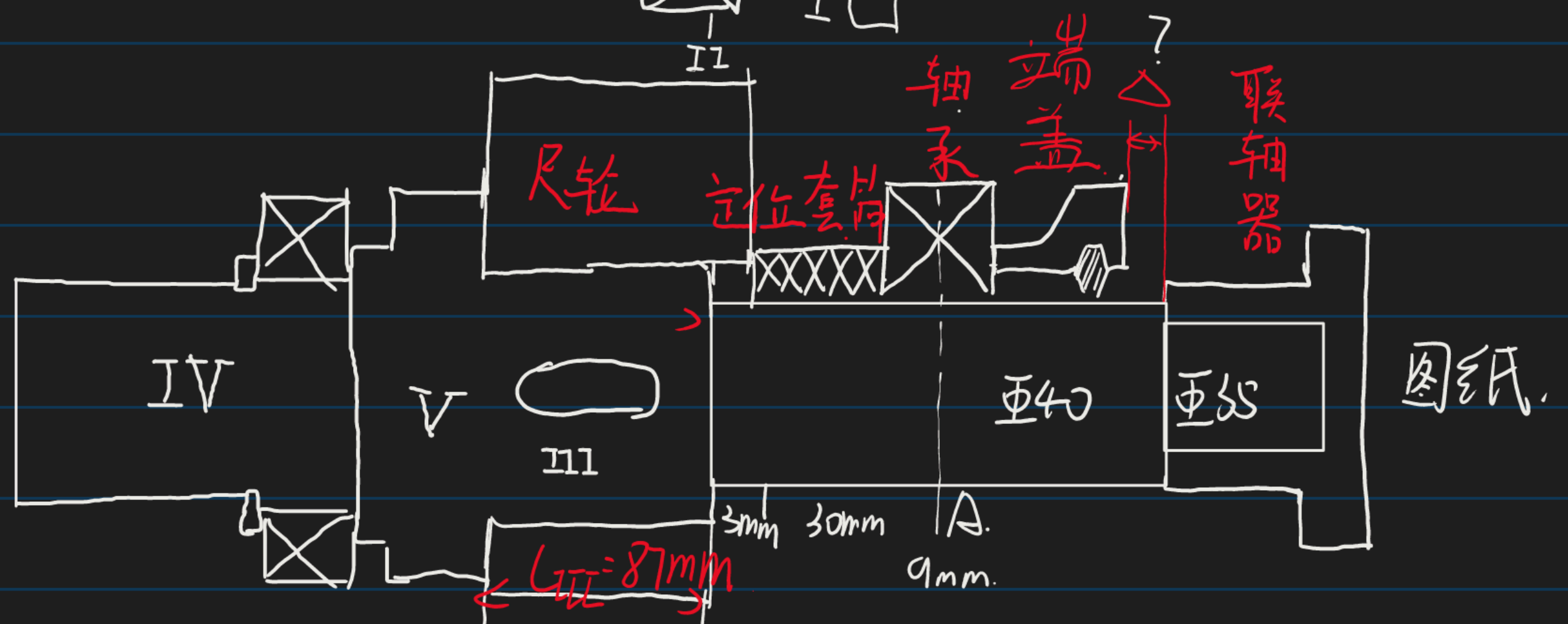
$P_2\rightarrow P_1\cdot\eta=4\cdot.9=3.9kW$

$n_2\rightarrow 130r/min$

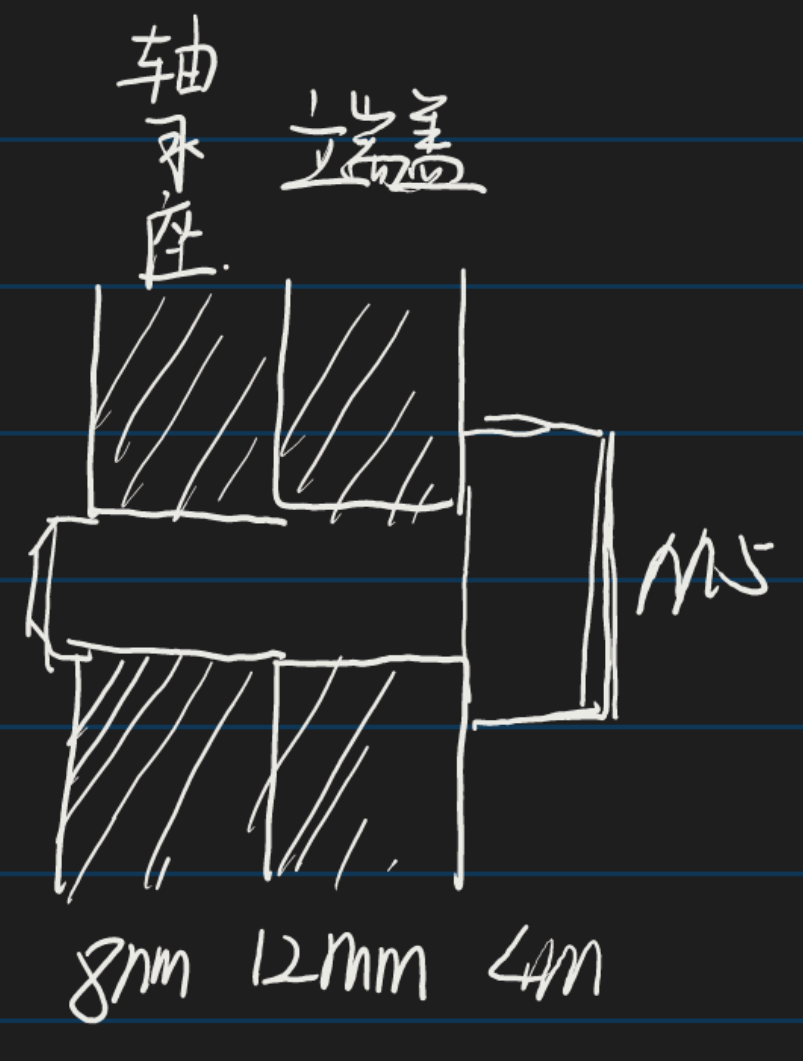
$d_{min}=112\cdot\sqrt[3]{\frac{3.9}{130}}=112\times0.31=34.72mm$

③





II
R轮
定位套筒
轴承
端盖
?
联轴器
IV
V
III
亚40
亚35
图纸.
3mm 30mm
A
9mm.
L_III=87mm

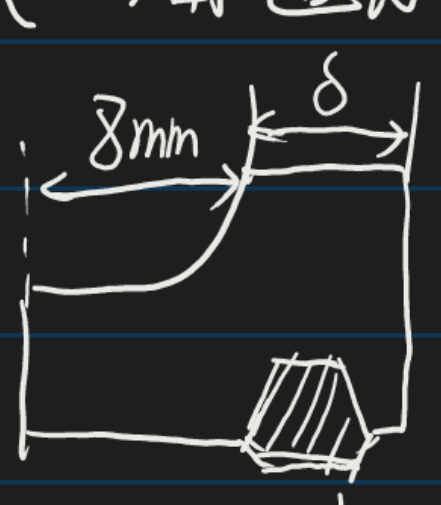


轴承座
端盖
M5
8mm 12mm 4m

I { 联轴器 { l 长度 60
d 内径 35mm. ⇒ J35×60 LT7.
装配结构. ⇔ 平键连接.

II { 轴承 ⇒ L_{II} d_{II} 长 宽 内径40.
定位元件 → { Fr=729N, Ft=1959N } $\sqrt{F_r^2+F_t^2}$ 6208深沟球轴承
密封. Fa=416N. Fa=416N

油毡封圈. { 内径, 线速度 定位套筒 { 内径40mm. 外径80mm. 长度30mm. 螺栓 { 粗M5, Δ: 4mm., 轴承座厚8mm

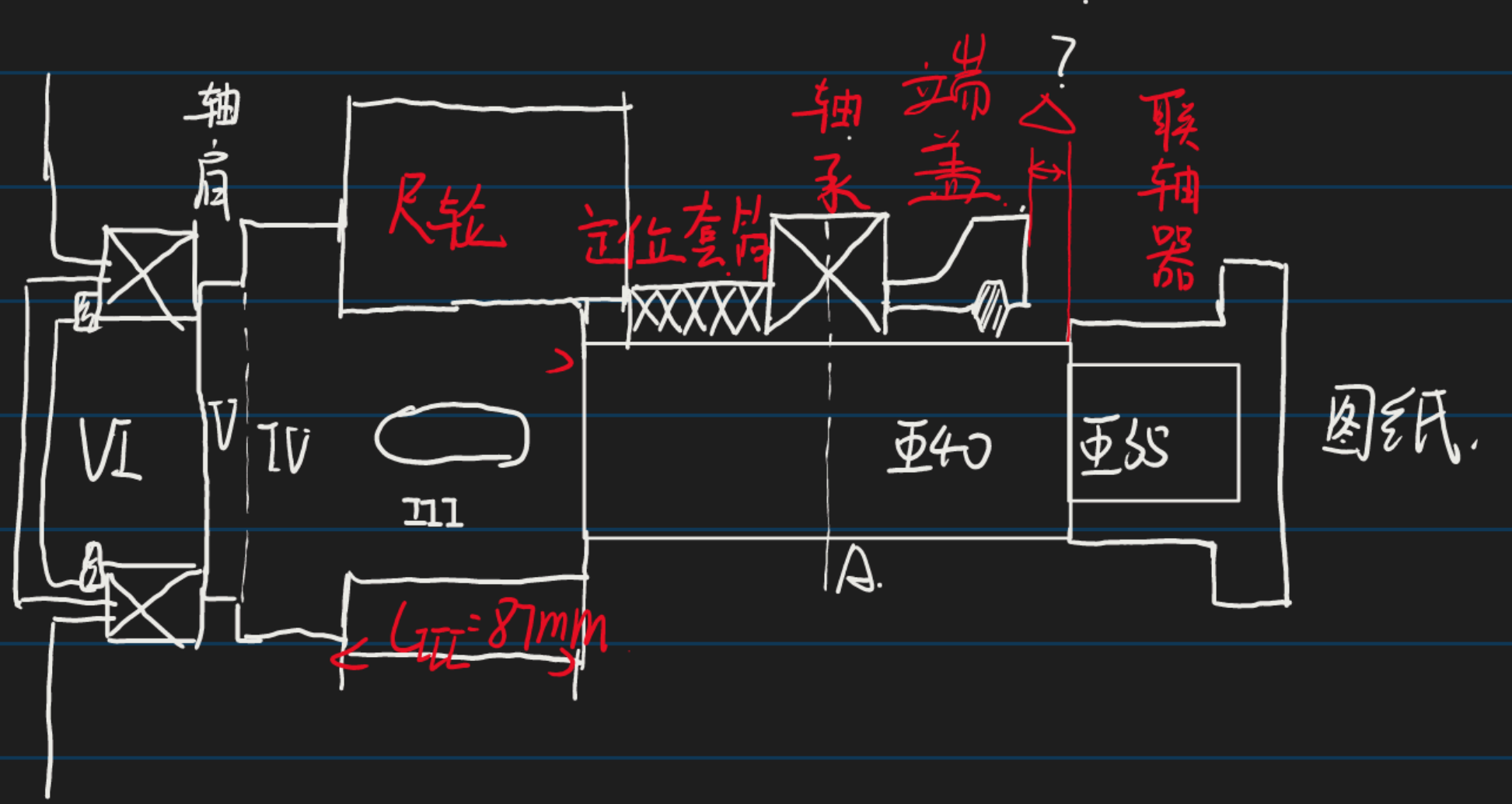


油封. 端盖20mm ∴ $L_{II}=4+12+8+18+30=72mm+3mm=75mm$

↑缝

D3=45mm

III { 齿轮: d_{III} 从d_{II}过渡.
扭矩 → 键 → 轴. 扭矩为286.6N·m. 轴亚45mm 键14×9×50
$L_{III}=87mm$



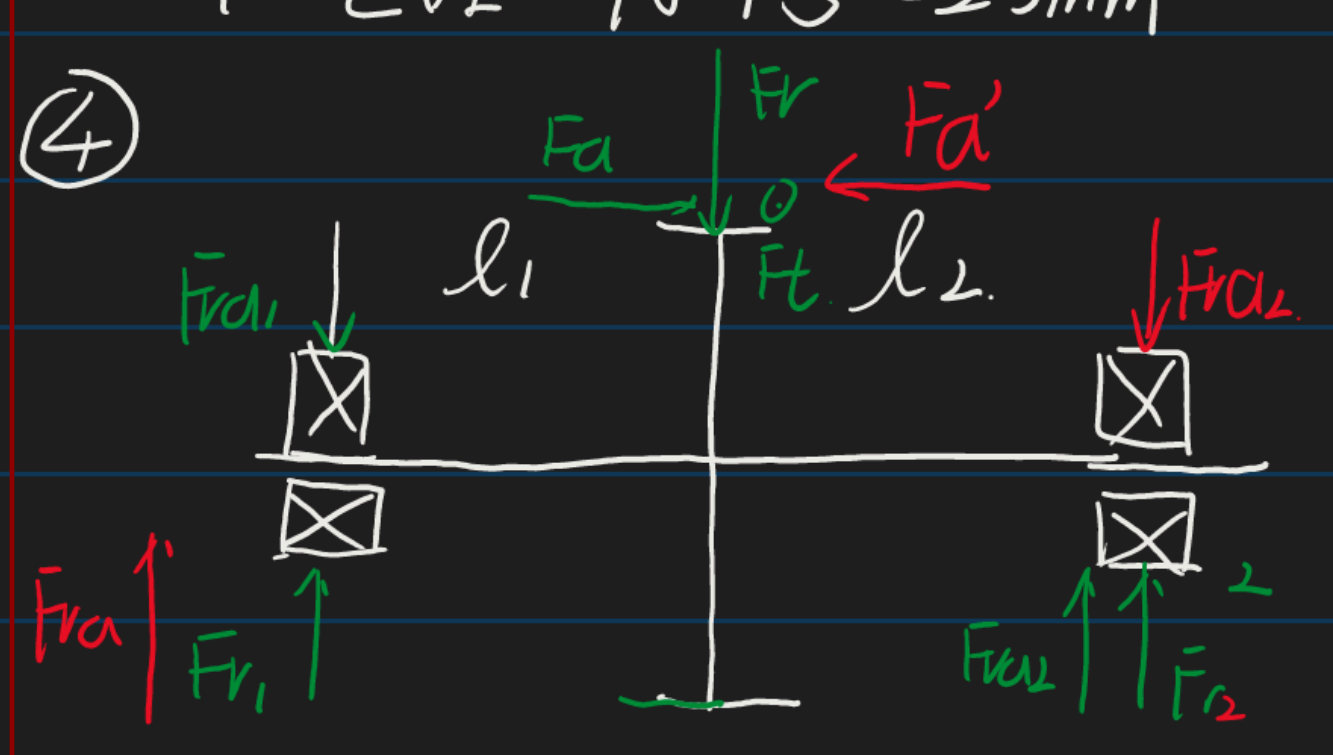
轴肩
R轮
定位套筒
轴承
端盖
?
联轴器
VI
V
IV
III
亚40
亚35
图纸.
A
L_III=87mm

IV → d_{IV} L_{IV} $d_{IV}=50mm$ $L_{IV}=10mm$

V { $d_V=40、50=48mm$
$L_V=3mm$ 轴肩

VI { 轴承: 18mm长 轴端到挡 ≥2.1mm → 2.5mm
挡 2.5mm } 5mm
$L_{VI}=18+5=23mm$

④



Fa
Fr
Fa'
Ft
l1
l2
Fra1
Fra2
Fra
Fr1
Fra2
Fr2
2

据③得到的数值.

$l_1=9+3+10+45=67$

$l_2=45+30+9=84$.

$F_{ra1}=F_{ra2}=F_{ra}$

由于$F_{r1}>F_{r2}$ ∴ 选左 $F_{r1}-F_{ra}$ 和右 $F_{r2}+F_{r2}$ 负载更均匀

目标: 求 { F_{r1}, F_{r2} } { F_{ra1}, F_{ra2} } { $F_{r1}+F_{r2}=F_r$, $F_{r1}l_1=F_{r2}l_2$ } { $F_{ra}=F_{ra1}=F_{ra2}$, $F_{ra}\cdot\frac{1}{2}(l_1+l_2)=F_a\cdot\frac{d_2}{2}$ } Fr=729N Fa=416N $d_2=300mm$

$F'=F_a\cdot\frac{1}{2}d=416\times0.15=62.4N\cdot m$

$F_{rV}=F_{re}\times0.67-62.4N/0.67+0.84=282N$

$F_{r2V}=729-282=447N$

$F_{H1}=\frac{F_t\times67}{67+84}=869.2N$. $F_{r2}=\sqrt{1084.2^2+447^2}=1177.9N=1178N$

$F_{H2}=1959-F_{H1}=1089.2N$

⑤轴承: 6208轴承

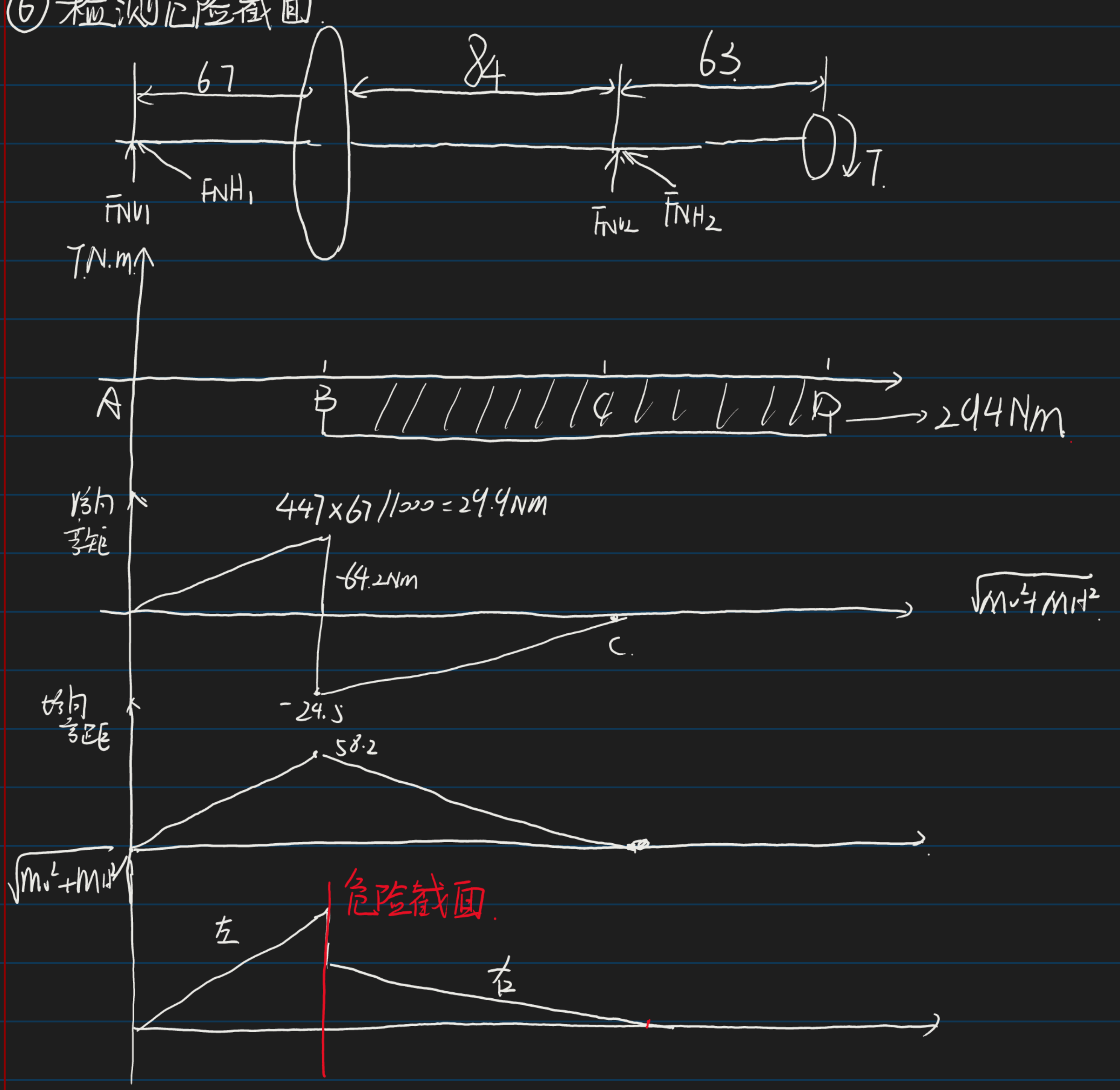
{ Fa=208N, Fr=1178N. } { $P=f_d(XF_r+YF_a)$ (f_d → 1.8), $L_h=\frac{10^6}{60n}(\frac{C}{P})^\varepsilon\geq L'_h$ (n → 130r/min, L'_h → 8000h) }

$F_a/C_0=208/18000=0.011$ $P=1.8\times1178N$

$F_a/F_r=0.17<e$ $L_h=\frac{10^6}{60n_m}(\frac{C}{P_m})^\varepsilon$ → [illegible]

{ X=1, Y=0.

⑥检测危险截面.



67
84
63
FNV1
FNH1
FNV2
FNH2
T.
T N.m
A
B
C
D
294Nm
竖向弯矩
447×67/1000=29.9Nm
-64.2Nm
C.
-24.5
水平弯矩
58.2
$\sqrt{M_V^2+M_H^2}$
$\sqrt{M_V^2+M_H^2}$
危险截面.
左
右

⑦弯扭合成校核轴的强度.

⑧键 槽选择原因.